

Mass-only obstructions for Abel-weighted Brownian responses

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Abstract

We study a two-channel Brownian response built from the actual von Mangoldt coefficients and a matched continuous Abel source. For every fixed $0 < \sigma < 1$, every sufficiently large real Y , and every finite common cutoff $N \geq Y$, we prove $J_4(Y, N) \gg_\sigma Y^{-3} \log Y$. The contribution of a single frequency band of size comparable to Y already gives this bound. The proof retains the mass-centering term, applies a local-gap mean-value theorem to the actual prime coefficients, and controls the continuous response by bounded variation. An elementary isolated-prime proof is retained as a weaker baseline requiring fewer external inputs. The Mellin transform of the uncut mass discrepancy, together with a self-contained Landau argument, supplies arbitrarily large mass zeros whenever the Riemann zeta function has a zero to the right of $\Re s = \sigma$. Truncating at those scales leaves a small mass discrepancy but a polynomially large response. Under this zero hypothesis, for continuous nondecreasing cutoff functions a signed PNT tail estimate excludes $J_4 = O(|\mu|^\kappa)$ whenever $N/Y \geq (3/\kappa) \log Y - b\sqrt{\log Y}$, where $0 \leq b < 0.8476836$. In particular, the quartic budget fails for $N = \lfloor cY \log Y \rfloor$, $c \geq 3/4$. For the regular superlogarithmic cutoff class $N/(Y \log Y) \rightarrow \infty$, no positive power of the mass discrepancy uniformly bounds the response over all sufficiently large real Abel scales when $0 < \sigma < 1/2$. At $\sigma = 1/2$, the corresponding uniform budget would imply RH; no converse is asserted. The result does not exclude an estimate along a prescribed dyadic schedule or a selected cofinal subsequence.

1 The response and the main result

Status convention. The theorems and lemmas below are internally proved statements [T]; their no-go consequences are [N]. External classical inputs are [R], finite computations are [E], and the remaining arithmetic and literature-priority questions are [O].

Fix $0 < \sigma < 1$. Put $a = 1 - \sigma$. For $Y > 0$ and $1 < N \leq \infty$, define positive measures on the nonnegative lag axis by

$$\begin{aligned} \alpha_{Y,N} &= \sum_{2 \leq n \leq N} \Lambda(n) n^{-\sigma} e^{-n/Y} \delta_{\log n}, \\ d\beta_{Y,N}(\lambda) &= \mathbf{1}_{[0, \log N]}(\lambda) e^{a\lambda - e^\lambda/Y} d\lambda. \end{aligned} \tag{1}$$

The upper endpoint is omitted when $N = \infty$. Write

$$A = \alpha_{Y,N}(\mathbb{R}), \quad B = \beta_{Y,N}(\mathbb{R}), \quad S = A + B, \quad M = A - B, \quad \mu = M/S.$$

For a measure on the nonnegative axis let η^{ev} denote its even symmetrization, with half the mass reflected to each side, and set

$$p = \alpha_{Y,N}^{\text{ev}} - A\delta_0, \quad c = B\delta_0 - \beta_{Y,N}^{\text{ev}}, \quad r = p + c.$$

For a real signed measure η of total mass zero, let $F_\eta(t) = \eta((-\infty, t])$. Our normalization is

$$\begin{aligned} D &= \|F_p\|_2^2 + \|F_c\|_2^2, \\ P^2 &= \|F_{r*r*p}\|_2^2 + \|F_{r*r*c}\|_2^2, \quad J_4(Y, N) = \frac{P^2}{S^4 D}. \end{aligned} \tag{2}$$

All these quantities are finite for the sources above, and $S, D > 0$.

The strongest bounds are proved in Theorems 7.2 and 8.2 below. They use classical mean-value and PNT inputs. We first give the elementary baseline, which exposes the mass-root mechanism without these additional estimates.

Theorem 1.1. *Put $L = \log Y$ and $N(Y) = \lfloor YL^2 \rfloor$.*

1. *For every fixed $0 < \sigma < 1$, uniformly for sufficiently large real Y ,*

$$J_4(Y, N(Y)) \geq c_\sigma Y^{-9} L^{-7} \tag{3}$$

with $c_\sigma > 0$.

2. *If ζ has a nonreal zero ρ with $\Re \rho > \sigma$, then for every $\kappa > 0$ and every sufficiently large Y_0 ,*

$$\sup_{\substack{Y \geq Y_0 \\ \mu(Y, N(Y)) \neq 0}} \frac{J_4(Y, N(Y))}{|\mu(Y, N(Y))|^\kappa} = \infty. \tag{4}$$

Corollary 1.2. *Conclusion (4) holds unconditionally for $0 < \sigma < 1/2$. For any $0 < \sigma < 1$, an eventual bound $J_4(Y, N(Y)) \leq C|\mu(Y, N(Y))|^4$ on all real scales with nonzero mass discrepancy implies that ζ has no zero in $\Re s > \sigma$. In particular, at $\sigma = 1/2$ it implies RH.*

Only the existence of a critical-line zero is needed for the first assertion of the corollary; this is a classical theorem [1]. The no-go concerns the mass-only budget in (2). It is neither a disproof of RH nor an obstruction to every realization of a Weil-type configuration.

2 Brownian energy and a positive actual response

We use the Fourier convention $\hat{\eta}(\xi) = \int_{\mathbb{R}} e^{-i\xi x} d\eta(x)$.

Lemma 2.1. *For a real signed measure η of mass zero and finite first absolute moment,*

$$\|F_\eta\|_2^2 = -\frac{1}{2} \iint |x - y| d\eta(x) d\eta(y) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|\hat{\eta}(\xi)|^2}{\xi^2} d\xi. \tag{5}$$

For $k_\lambda = (\delta_\lambda + \delta_{-\lambda})/2 - \delta_0$,

$$\langle F_{k_\lambda}, F_{k_\gamma} \rangle = \frac{1}{2} \min(\lambda, \gamma) \quad (\lambda, \gamma \geq 0).$$

Proof. The squared difference of two half-line indicators has integral $|x - y|$. Expanding its integral against $\eta \otimes \eta$, the two single-variable terms vanish because $\eta(\mathbb{R}) = 0$. The Fourier formula follows from $F'_\eta = \eta$ and Plancherel. Finite first absolute moment justifies the distance integral. For the final identity, each primitive is constant with magnitude $1/2$ on each half of $[-\lambda, \lambda]$, with opposite signs on the two halves. Their overlap has length $2 \min(\lambda, \gamma)$. $\square$

Lemma 2.2. *For every $Y > 0$, the uncut response $J_4(Y, \infty)$ is strictly positive and continuous in Y .*

Proof. The measure r has a nonzero atom at $\log 2$. The continuous source has no atom there, so $r \neq 0$. Fourier uniqueness gives some $\xi \neq 0$ with $\widehat{r}(\xi) \neq 0$. For every real $\xi \neq 0$,

$$\widehat{c}(\xi) = \int_1^\infty x^{-\sigma} e^{-x/Y} (1 - \cos(\xi \log x)) dx > 0.$$

Thus $\widehat{r * r * c} = \widehat{r}^2 \widehat{c}$ is not identically zero, and Lemma 2.1 gives strictly positive energy. On any compact Y -interval the source coefficients and their first absolute logarithmic moments admit a common integrable majorant. They vary continuously in the corresponding weighted total-variation norm. Convolution and the distance integral preserve this continuity, proving the assertion. $\square$

3 Mellin poles force mass cancellation

Write

$$M_\infty(Y) = \sum_{n \geq 2} \Lambda(n) n^{-\sigma} e^{-n/Y} - \int_1^\infty x^{-\sigma} e^{-x/Y} dx.$$

Lemma 3.1. *For $\Re s > 1 - \sigma$,*

$$\int_0^\infty M_\infty(Y) Y^{-s-1} dY = \Gamma(s) \left(-\frac{\zeta'}{\zeta}(\sigma + s) - \frac{1}{\sigma + s - 1} \right). \quad (6)$$

The meromorphic function on the right is holomorphic near every positive real point. At $s = \rho - \sigma$, for a nonreal zero ρ of multiplicity m_ρ , its residue is $-m_\rho \Gamma(\rho - \sigma) \neq 0$ whenever $\Re \rho > \sigma$.

Proof. The bound $\Lambda(n) \leq \log n$ gives $A_\infty + B_\infty \ll_\sigma Y^{1-\sigma} (1 + \log Y)$ for $Y \geq 2$; at zero both masses decay exponentially in $1/Y$. Absolute convergence therefore permits interchange of integrals and sums. The substitution $t = x/Y$ gives $\int_0^\infty e^{-x/Y} Y^{-s-1} dY = \Gamma(s) x^{-s}$. The Euler logarithmic derivative and the continuous integral yield (6).

There are no positive real zeros of ζ : for $w > 1$ use its Euler product, and for $0 < w < 1$ use

$$\zeta(w) = \frac{\sum_{n \geq 1} (-1)^{n-1} n^{-w}}{1 - 2^{1-w}} < 0.$$

At $w = 1$ the two displayed pole parts cancel. The Gamma function is holomorphic on the positive real axis. At a nonreal zero, the negative logarithmic derivative has residue $-m_\rho$, and Gamma has no zero, establishing the last assertion. $\square$

We include the classical positivity argument to specify the analytic continuation required in the proof. A modern statement of this Landau mechanism appears in [3, Theorem 3.1]; we make no novelty claim for the mechanism.

Lemma 3.2 (Landau mechanism). *Suppose $f(Y) \geq 0$ for $Y \geq Y_0 \geq 1$, and the Mellin integral $G(s) = \int_{Y_0}^\infty f(Y) Y^{-s-1} dY$ has finite convergence abscissa c . Then G cannot continue holomorphically across the real point c .*

Proof. If it could, choose a small $\varepsilon > 0$ such that the Taylor radius at $b = c + \varepsilon$ exceeds ε . For every integer $k \geq 0$,

$$(-1)^k G^{(k)}(b) = \int_{Y_0}^\infty f(Y) (\log Y)^k Y^{-b-1} dY \geq 0.$$

Choose $h > \varepsilon$ within that Taylor radius. Taylor convergence and Tonelli imply

$$\int_{Y_0}^\infty f(Y) Y^{-(b-h)-1} dY = \sum_{k \geq 0} \frac{h^k}{k!} (-1)^k G^{(k)}(b) < \infty,$$

contrary to $b - h < c$. $\square$

Theorem 3.3. *If ζ has a nonreal zero ρ with $\Re \rho > \sigma$, then M_∞ takes both strictly positive and strictly negative values arbitrarily far out. In particular, it has a sequence of zeros $Y_j \rightarrow \infty$.*

Proof. Assume first that M_∞ is eventually nonnegative. Its nonnegative tail Mellin transform has convergence abscissa $c \leq 1 - \sigma$. Removing a bounded initial Y -interval from (6) subtracts an entire function, because of the exponential decay at zero. If $c < \Re \rho - \sigma$, the tail integral would be holomorphic at $\rho - \sigma$, contradicting the pole there and uniqueness of meromorphic continuation. Hence $c \geq \Re \rho - \sigma > 0$. Lemma 3.2 now contradicts holomorphy at the positive real point c . Apply the same argument to $-M_\infty$ to exclude eventual nonpositivity. The mass is real analytic for $Y > 0$, by locally dominated differentiation of its defining sums and integrals. The intermediate value theorem supplies the stated zeros. $\square$

By Lemma 2.2, even omitting all exact mass zeros cannot rescue a uniform uncut mass-only bound: the response stays positive while the mass tends to zero near each root. The next section proves a quantitative version that survives a finite arithmetic cutoff.

4 An isolated prime atom survives the continuous background

Proof of Theorem 1.1(1). Take $L = \log Y$, $N = \lfloor YL^2 \rfloor$. Bertrand's theorem [2] provides a prime $q \in (Y, 2Y)$; apply its integer form at $\lfloor Y \rfloor$. For sufficiently large Y , we have $q \leq N < q^2$. Put $d = \alpha^{\text{ev}} - M\delta_0$ and $b = \beta^{\text{ev}}$, so $r = d - b$. The atomic part of $\eta = r * r * p$ is $d * d * p$. Its remaining part

$$g(x) dx = -2d * b * p + b * b * p$$

is absolutely continuous.

At $x_0 = 3 \log q$ the atomic coefficient is exactly

$$w = \frac{1}{8} ((\log q) q^{-\sigma} e^{-q/Y})^3 \gg_\sigma L^3 Y^{-3\sigma}. \quad (7)$$

Indeed every atomic term uses at most three signed prime-power lags. Since $q^2 > N$, each lag contributes at most one to the q -adic valuation. A total valuation of three requires three positive lags divisible by q ; their product is q^3 , forcing each factor to equal q . Neither a center term nor a negative lag can participate. An absolutely continuous term cannot alter this atomic coefficient.

Every other atomic location is $\log(u/v)$ for positive integers $u, v \leq N^3$. If $u/v \neq q^3$, the distinct integers u, vq^3 have smaller member at most N^3 , and therefore

$$|\log(u/v) - 3 \log q| \geq \log(1 + N^{-3}) \geq (2N^3)^{-1}. \quad (8)$$

Set $\delta = (8N^3)^{-1}$, which isolates the atom on both sides.

Elementary summation and maximization of the continuous density give

$$S \ll_\sigma Y^{1-\sigma} L, \quad \left\| \frac{db}{dx} \right\|_\infty \ll_\sigma Y^{1-\sigma}.$$

Here x denotes the additive lag coordinate. Since $\|d\|_{\text{TV}} \leq 2S$, $\|p\|_{\text{TV}} \leq 2A$, and $\|b\|_{\text{TV}} = B$,

$$\|g\|_\infty \leq 10S^2 \left\| \frac{db}{dx} \right\|_\infty \ll_\sigma Y^{3(1-\sigma)} L^2. \quad (9)$$

Equations (7)–(9) imply $\delta \|g\|_\infty / w \ll_\sigma L^{-7}$, eventually at most $1/4$.

The two one-sided limits of F_η at x_0 differ by w ; one therefore has absolute value at least $w/2$. On the corresponding adjacent interval of length δ there are no other atoms, and the continuous change is at most $w/4$. Consequently

$$P^2 \geq \|F_\eta\|_2^2 \geq \delta w^2 / 16 \gg_\sigma Y^{-6\sigma-3}.$$

Lemma 2.1 also gives

$$D \leq \frac{1}{2}(\log N)(A^2 + B^2), \quad S^4 D \ll_\sigma Y^{6(1-\sigma)} L^7.$$

Dividing yields (3). This proof permits the jumps of the integer-valued cutoff $N(Y)$. $\square$

5 Finite-cutoff mass cancellation and the obstruction

Proof of Theorem 1.1(2). Choose the uncut mass zeros Y_j from Theorem 3.3, and set $L_j = \log Y_j$, $N_j = \lfloor Y_j L_j^2 \rfloor$. At each such point the finite mass is the negative of a difference of omitted tails, so

$$|M(Y_j, N_j)| \ll_\sigma Y_j^{1-\sigma} L_j^2 e^{-L_j^2/2}. \quad (10)$$

For completeness, bound the prime tail by the integral of $(\log x)x^{-\sigma}e^{-x/Y}$ from N , which is decreasing there. After $x = Yu$, use $u^{-\sigma} \leq 1$ and $\log u \leq u$ for $u \geq 1$. The resulting upper bound is

$$C_\sigma Y^{1-\sigma} (L + N/Y + 1) e^{-N/Y}.$$

The continuous tail is smaller. This proves (10); the floor error in N/Y is harmless. The finite continuous mass on $[Y/2, Y]$ is $\gg_\sigma Y^{1-\sigma}$. Hence

$$|\mu(Y_j, N_j)| \ll_\sigma L_j^2 e^{-L_j^2/2}.$$

To ensure nonzero finite mass, perturb Y_j slightly to the right. The cutoff is constant on a right neighborhood, even when Y_j is a cutoff threshold. For fixed N_j , the function $M(Y, N_j)$ is real analytic and not identically zero: at small Y its integral beginning at 1 dominates the prime terms beginning at 2. Its zeros are therefore isolated. Choose $0 < Y'_j - Y_j < Y_j^{-1}$ within the same cutoff cell, with $M(Y'_j, N_j) \neq 0$, and small enough to retain (10) up to a constant.

Theorem 1.1(1) applies directly to these finite sources; no response-tail transfer is required. For $\kappa = 4$,

$$\frac{J_4(Y'_j, N_j)}{|\mu(Y'_j, N_j)|^4} \gg_\sigma e^{2L_j^2 - 9L_j} L_j^{-15} \longrightarrow \infty.$$

For general $\kappa > 0$, the exponential factor is $e^{\kappa L_j^2/2 - 9L_j}$, which still dominates all logarithmic factors. This proves (4). $\square$

Proof of Corollary 1.2. For $0 < \sigma < 1/2$, a classical nonreal critical-line zero has real part exceeding σ , so apply Theorem 1.1(2). In general, any zero to the right of σ contradicts the proposed uniform budget. At $\sigma = 1/2$, the symmetry of the nontrivial zeros about the critical line then gives RH. $\square$

6 Uniformity across a class of cutoff rules

Theorem 6.1. *For every fixed $0 < \sigma < 1$, there are $c_\sigma > 0, Y_\sigma$ such that, simultaneously for every real $Y \geq Y_\sigma$ and every integer $Y \leq N \leq Y^2/8$,*

$$J_4(Y, N) \geq c_\sigma Y^{-6} N^{-3} (\log Y)^{-1}. \quad (11)$$

The constant and the threshold are independent of N .

Proof. This time apply Bertrand's theorem at $\lfloor Y/2 \rfloor$ to choose $Y/2 < q < Y$. Then $q \leq N < q^2$. The valuation argument and integer separation from Section 4 apply without change: the atom at $3 \log q$ has coefficient $w \gg_\sigma L^3 Y^{-3\sigma}$ and can be isolated with $\delta = (8N^3)^{-1}$. The uncut source bounds give, uniformly in the allowed N ,

$$S \ll_\sigma Y^{1-\sigma} L, \quad \|g\|_\infty \ll_\sigma Y^{3(1-\sigma)} L^2, \quad \frac{\delta \|g\|_\infty}{w} \ll_\sigma \frac{Y^3}{N^3 L} \leq \frac{C_\sigma}{L}.$$

For a threshold independent of N , the jump argument therefore gives

$$P^2 \geq \delta w^2 / 16 \gg_\sigma N^{-3} L^6 Y^{-6\sigma}.$$

Since $\log N \leq 2L$, the denominator satisfies $S^4 D \ll_\sigma Y^{6(1-\sigma)} L^7$. Dividing proves (11). Only $\log N \asymp \log Y$, not asymptotic equality, is used. $\square$

Theorem 6.2. *Let Φ be continuous and nondecreasing on a final real interval, and put $N(Y) = \lfloor \Phi(Y) \rfloor$. Assume eventually that*

$$Y \leq N(Y) \leq Y^2/8, \quad \frac{N(Y)}{Y \log Y} \rightarrow \infty.$$

If zeta has a nonreal zero to the right of σ , then (4) holds for this cutoff rule, for every fixed $\kappa > 0$ and every sufficiently large Y_0 .

Proof. Take the uncut mass roots Y_j from Theorem 3.3, and put $h_j = N(Y_j)/Y_j$. The same decreasing-integral estimate used in (10), now with lower limit h_j , gives

$$|M(Y_j, N_j)| \ll_\sigma Y_j^{1-\sigma} (L_j + h_j + 1) e^{-h_j}.$$

The continuum mass on $[Y_j/2, Y_j]$ bounds S below. Since $h_j/L_j \rightarrow \infty$, this yields

$$|\mu(Y_j, N_j)| \ll_\sigma (L_j + h_j + 1) e^{-h_j} \leq e^{-h_j/2}$$

eventually. Continuity and monotonicity of Φ imply that $N(Y) = N_j$ in a right neighborhood of Y_j : continuity keeps $\Phi(Y) < N_j + 1$, and monotonicity keeps $\Phi(Y) \geq N_j$. This includes the case of an integer value $\Phi(Y_j)$. The fixed- N_j analytic perturbation from Section 5 therefore provides nonzero mass points Y'_j , with $0 < Y'_j - Y_j < Y_j^{-1}$, retaining the bound up to a factor of two.

Theorem 6.1 and $N_j \leq (Y'_j)^2/8$ give $J_4(Y'_j, N_j) \gg_\sigma (Y'_j)^{-12} (\log Y'_j)^{-1}$. Consequently

$$\frac{J_4(Y'_j, N_j)}{|\mu(Y'_j, N_j)|^\kappa} \gg_{\sigma, \kappa} L_j^{-1} \exp(\kappa h_j/2 - 12L_j) \rightarrow \infty.$$

$\square$

This includes each fixed cutoff $N = \lfloor Y(\log Y)^b \rfloor$ with $b > 1$, and $N = \lfloor Y^{1+\varepsilon} \rfloor$ with $0 < \varepsilon < 1$. For any selected discrete schedule within $Y \leq N \leq Y^2/8$, the budget $J_4 \leq C|\mu|^\kappa$ at nonzero mass points necessarily requires

$$|\mu| \geq (c_\sigma/C)^{1/\kappa} Y^{-6/\kappa} N^{-3/\kappa} (\log Y)^{-1/\kappa}.$$

This is not sufficient for the budget, and no violation of this necessary condition on a prescribed dyadic schedule is proved.

7 A cutoff-uniform mesoscopic response floor

All cutoffs in this section are finite, but no upper bound on N/Y is imposed. The constants may depend on fixed $0 < \sigma < 1$. Set $L = \log Y$, $a = 1 - \sigma$, and

$$a_n = \mathbf{1}_{n \leq N} \Lambda(n) n^{-\sigma} e^{-n/Y}, \quad F(\xi) = \sum_n a_n n^{-i\xi},$$

$$C(\xi) = \int_1^N x^{-\sigma-i\xi} e^{-x/Y} dx, \quad Q = \sum_n a_n^2, \quad Q_1 = \sum_n n a_n^2.$$

The exact centered symbols are

$$\hat{p} = \Re F - A, \quad \hat{c} = B - \Re C, \quad \hat{r} = \Re F - M - \Re C.$$

In particular the term $-M$ cannot be discarded at large frequencies.

Lemma 7.1. *Uniformly for finite $N \geq Y$,*

$$S \asymp_\sigma Y^a, \quad D \asymp_\sigma L S^2, \quad Q_1 \ll_\sigma Y^{2-2\sigma} L, \quad Q \gg_\sigma Y^{1-2\sigma} L, \quad Q_1 \ll_\sigma Y Q.$$

Proof. Chebyshev's upper bound, split into dyadic blocks with the Abel weight, gives

$$A \ll_\sigma Y^a, \quad B \asymp_\sigma Y^a, \quad \int \lambda d\alpha + \int \lambda d\beta \ll_\sigma Y^a L.$$

The sums below Y converge geometrically since $a > 0$; those above Y converge exponentially. Positive terms permit a bound by the uncut sums, independently of $\log N$. Since $\Lambda(n)^2 \leq \Lambda(n) \log n$, the same argument gives the asserted Q_1 bound; its required exponent is $2 - 2\sigma > 0$. Qualitative PNT on primes $p \in (Y/2, Y]$ gives the lower bound for Q , using $(\log p)^2 \geq \log(Y/2) \log p$. The minimum-kernel identity and $\min(\lambda, \nu) \leq \lambda$ give $D \ll_\sigma L Y^{2a}$. The continuous mass on $[Y/2, Y]$ alone gives the reverse inequality. $\square$

Theorem 7.2. *There are $K_\sigma, c_\sigma, Y_\sigma > 0$ such that for every real $Y \geq Y_\sigma$ and finite integer $N \geq Y$, with $U = K_\sigma Y$ and $I = [U, 2U]$, the contribution from the single positive-frequency band I satisfies*

$$J_{4,I} \geq c_\sigma \frac{(Q + M^2)^2}{Y L S^4} \geq c_\sigma \frac{\log Y}{Y^3}. \quad (12)$$

In particular $J_4 \geq J_{4,I}$.

Proof. For finite distinct real frequencies λ_j , put $\delta_j = \min_{k \neq j} |\lambda_j - \lambda_k|$. The local-gap mean-value formula of Montgomery and Vaughan [4, Corollary 2, (1.9)] gives

$$\int_V^{V+U} \left| \sum_j d_j e^{i\lambda_j t} \right|^2 dt = U \sum_j |d_j|^2 + O \left(\sum_j \delta_j^{-1} |d_j|^2 \right).$$

The constant is absolute, and translation changes only coefficient phases. Apply this to

$$f(\xi) = \Re F(\xi) - M = -M + \frac{1}{2} \sum_n a_n (e^{i\xi \log n} + e^{-i\xi \log n}).$$

The frequency set $\{0, \pm \log n : a_n > 0\}$ satisfies $\delta_0 = \log 2$ and $\delta_{\pm \log n} \geq 1/(n+1) \geq 1/(2n)$. Thus, retaining the zero-frequency mass coefficient,

$$\int_U^{2U} f(\xi)^2 d\xi = U(M^2 + Q/2) + O(Q_1 + M^2).$$

Lemma 7.1 allows a fixed K_σ to absorb this error, so the integral is at least $cU(Q + M^2)$.

The continuous lag density is unimodal and has maximum $O_\sigma(Y^a)$. After restriction to its interval and zero extension, its total variation is at most twice this maximum. Integration by parts, including both endpoint jumps, gives

$$|C(\xi)| \ll_\sigma Y^a/|\xi|, \quad \int_I |C(\xi)|^2 d\xi \ll_\sigma Y^{2a}/U.$$

The latter is $o(UQ)$, since its ratio to UQ is $O_\sigma((Y \log Y)^{-1})$. The reverse triangle inequality in $L^2(I)$ therefore gives

$$\int_I |\hat{r}|^2 d\xi \gg_\sigma U(Q + M^2).$$

Also $\hat{c} = B - \Re C \geq B/2$ throughout I , eventually. Cauchy's inequality now implies $\int_I |\hat{r}|^4 d\xi \gg_\sigma U(Q + M^2)^2$. Retaining the final continuous physical direction, we obtain

$$J_{4,I} = \frac{1}{2\pi S^4 D} \int_I \frac{|\hat{r}|^4(|\hat{p}|^2 + |\hat{c}|^2)}{\xi^2} d\xi \geq \frac{B^2}{32\pi U^2 S^4 D} \int_I |\hat{r}|^4 d\xi.$$

Use the scale bounds in Lemma 7.1 to finish. No limiting assertion for infinite N is needed. $\square$

Corollary 7.3. *On any actual schedule with $M \neq 0$, the budget $J_4 \leq C\mu^4$ requires*

$$|M| \geq c_{\sigma,C} Y^{1/4-\sigma} (\log Y)^{1/4}.$$

If $M = o(Y^{1/4-\sigma} (\log Y)^{1/4})$ along a cofinal sequence of nonzero masses, then $J_4/\mu^4 \rightarrow \infty$.

Proof. Divide (12) by $\mu^4 = M^4/S^4$ and use the lower bound for Q . $\square$

This is a necessary condition, not a proof of failure on a dyadic sequence. In particular, a separate two-endpoint prime-jump selection guarantees only $|M| \gg Y^{-\sigma} \log Y$; this weaker guaranteed scale does not upper-bound the actual selected mass.

8 Signed tails and logarithmic cutoff rules

Write $c_* = 0.8476836$. We use the published quantitative PNT of Fiori, Kadiri and Swidinsky [5, Corollary 1.4] only through

$$|\psi(x) - x| \ll_d x e^{-d\sqrt{\log x}} \quad (0 < d < c_*, x \geq 2).$$

The logarithmic prefactor in the cited bound is absorbed by the strict inequality $d < c_*$. No claim of optimality for c_* is made here.

Lemma 8.1. *For fixed $0 < \sigma < 1$, $0 < d < c_*$, and $h = N/Y \geq 1$,*

$$|M_\infty(Y) - M(Y, N)| \ll_{\sigma,d} Y^{1-\sigma} h^{1-\sigma} e^{-h-d\sqrt{\log Y}},$$

uniformly for all finite integer $N \geq Y$.

Proof. Put $R(x) = \psi(x) - x$ and $w(x) = x^{-\sigma} e^{-x/Y}$. Matched right endpoints give the exact identity

$$M_\infty - M(Y, N) = -w(N)R(N) - \int_N^\infty R(x)w'(x) dx.$$

The value $R(N)$ is right-continuous, and the prime tail is strictly $n > N$. The boundary term at infinity vanishes. On $x \geq N \geq Y$, the PNT bound is at most $C_d x e^{-d\sqrt{\log Y}}$. Substitution $x = Yu$ bounds the absolute value by $C_d Y^a e^{-d\sqrt{\log Y}}$ times

$$h^a e^{-h} + \sigma \int_h^\infty u^{-\sigma} e^{-u} du + \int_h^\infty u^a e^{-u} du.$$

For $h \geq 1$, write $u = h + v$ and bound $(h + v)^a \leq h^a(1 + v)^a$. Both integrals are $O_\sigma(h^a e^{-h})$, as required. $\square$

Theorem 8.2. Fix $0 < \sigma < 1$, $\kappa > 0$, and $0 \leq b < c_*$. Let Φ be continuous and nondecreasing on a final real interval, and put $N(Y) = \lfloor \Phi(Y) \rfloor < \infty$. Suppose eventually $N(Y) \geq Y$ and

$$\frac{N(Y)}{Y} \geq \frac{3}{\kappa} \log Y - b\sqrt{\log Y}. \quad (13)$$

If zeta has a nonreal zero with real part greater than σ , then (4) holds for this κ and cutoff rule. There is no upper bound required on N/Y .

Proof. Choose $b < d < c_*$, and take the mass roots Y_j from Theorem 3.3. Put $L_j = \log Y_j$ and $h_j = N(Y_j)/Y_j$. Lemma 8.1 and $S \asymp_\sigma Y^a$ imply

$$|\mu(Y_j, N_j)| \ll_{\sigma, d} h_j^a e^{-h_j - d\sqrt{L_j}} \ll_{\sigma, d, \kappa} L_j^a e^{-3L_j/\kappa - (d-b)\sqrt{L_j}}.$$

For the second inequality use the monotonicity of $h^a e^{-h}$ on $h \geq 1$; no upper bound on h_j is being assumed. As in the preceding cutoff theorem, continuity and monotonicity provide a right-hand cell with fixed N_j . Nonconstant real analyticity then permits Y'_j in that cell with $0 < Y'_j - Y_j < Y_j^{-1}$, nonzero mass, and the same upper bound up to a fixed factor. Since $N_j/Y_j \rightarrow \infty$, also $N_j \geq Y'_j$. Theorem 7.2 gives

$$\frac{J_4(Y'_j, N_j)}{|\mu(Y'_j, N_j)|^\kappa} \gg_{\sigma, d, \kappa} L_j^{1-a\kappa} e^{\kappa(d-b)\sqrt{L_j}} \rightarrow \infty.$$

□

For $\kappa = 4$, every rule $N = \lfloor cY \log Y \rfloor$, $c \geq 3/4$, is covered. At $c = 3/4$, choose any fixed $b > 0$ to absorb rounding; one must not silently apply $b = 0$. The theorem is unconditional for $\sigma < 1/2$. At $\sigma = 1/2$, the corresponding continuous budget implies RH, without a converse. A fixed logarithmic cutoff need not exclude every positive exponent κ ; the condition (13) must be checked for that exponent. The boundary $3/\kappa$ is sufficient for this method, not claimed to be optimal. Dyadic selections with $N \asymp Y$ remain outside this obstruction.

9 Scope, the Weil interface, and reproducibility

The continuous-scale quantifier in the no-go theorems is essential. A bound for $Y = \theta 2^m$, simultaneously with the same constant and starting index for all $1 \leq \theta < 2$, would cover all large real scales and is subject to the theorem. A prescribed θ , constants depending on θ , or an existentially chosen cofinal schedule are not covered. In particular, a mass-relative estimate on a selected dyadic schedule remains an open arithmetic question.

The response (2) is the two-channel prime/continuous subsystem used in a conditional capture and Schur-gain criterion. It is not the complete Weil quadratic form. Incorporating the Gamma response, or constructing a cohomological realization, requires a separate bridge. The theorem only prevents treating a uniform continuous-scale mass-only budget as automatic positivity. The implication to RH at the central parameter is necessary, not an equivalence established here.

The elementary baseline uses the classical Euler identity, analytic continuation of zeta, existence of a critical-line zero, and Bertrand's theorem, but no PNT error estimate. The stronger mesoscopic floor additionally uses qualitative PNT and the local-gap mean-value theorem; the logarithmic cutoff extension uses the quantitative PNT stated above. None of these inputs assumes the location of unknown zeros. For other L -functions, positive real zeros or complex coefficients can invalidate the real-axis or positivity steps; no unrestricted GRH extension is claimed.

The accompanying exact script `scripts/abel_prime_atom_audit.py` checks multiplicative convolution coefficients and integer separation using rational arithmetic. The numerical script `scripts/abel_mass_discrepancy_probe.py` checks bounded actual Abel samples using numpy and mpmath. All samples at $\sigma = 1/4, 1/2, 3/4$, $Y = 2^3, \dots, 2^{18}$ were negative. This neither contradicts eventual oscillation nor estimates its first occurrence. The computations are not inputs to the asymptotic proofs. On the Windows test platform, numpy longdouble has the same 52 explicit mantissa bits as binary64. The printed tail majorants do not include floating-point coefficient or summation errors and are not certified total-error bounds. Independent internal audits checked the Mellin singularities, prime-atom lower bound, the cutoff-uniform local-gap mean value, continuous endpoint terms, cutoff transfer, and scale quantifiers. The additional script `scripts/prime_jump_full_response_probe.py` retains both actual channels, computes the full Brownian denominator independently, and checks finite frequency bands at selected dyadic cutoffs. These floating-point checks, including a separate 50-digit small case, establish no asymptotic budget or dyadic counterexample. The literature priority of their combination remains to be assessed.

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